



Shannon Entropy as a Forward Indicator of Volatility Regime Shifts in Energy Futures

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1 Abstract

This paper evaluates whether entropy-based metrics can provide earlier or more informative signals of risk regime changes in energy futures markets than conventional volatility models. Using daily returns from crude oil, natural gas, solar, and uranium over a multi-year sample, we systematically benchmarked rolling entropy against realized, EWMA, and GARCH volatility. The analysis assesses hit rates, false alarm rates, and parameter sensitivity across multiple bin and window choices. Results show that entropy sometimes signals regime changes before classical volatility statistics, but its predictive advantage is small, inconsistent, and frequently paired with high false alarm rates. For most assets and practical settings, traditional volatility metrics remain more reliable. Entropy is best used as a supplementary regime indicator within a broader risk management or systematic trading framework, not as a primary tool. These findings inform the design and evaluation of quantitative strategies for energy commodity trading, suggesting entropy-based methods add limited incremental value over established volatility models.

2 Introduction

Periods of extreme volatility in energy markets are difficult to anticipate. Geopolitical events, sudden supply disruptions, and shifts in market sentiment often result in large price swings with limited warning, even as standard risk measures such as realized volatility or GARCH forecasts remain stable. In practice, volatility spikes are most often recognized in hindsight, aftermarket impacts are already underway.

This research evaluates whether the entropy of daily returns can provide earlier signals of market instability than traditional volatility measures. Volatility is used to assess the dispersion of price changes, but entropy measures the unpredictability or randomness within the distribution of returns. At a given volatility level, higher entropy reflects a situation where outcomes are less concentrated and more widely dispersed. In this sense, entropy does not simply mirror the size of returns but captures evolving patterns within the distribution itself.

Entropy offers a distinct mathematical approach by responding to distributional complexity. For example, volatility may remain unchanged even if the underlying shape of returns shifts from a single cluster to multiple peaks or a broader spread. Entropy identifies such periods by detecting reductions in order and increases in randomness. This property is especially relevant during the transition to market instability, when conventional risk statistics fail to provide advance notice.

The main hypothesis being examined is whether increases in entropy consistently precede volatility spikes. Empirical analysis compares the timing and predictive value of entropy signals to those generated by rolling volatility, EWMA, and GARCH approaches. The inquiry centers on whether entropy contains additional information that can improve risk assessment and timing for participants in energy futures markets.

3 Methodology

3.1 Data and Market Context

We mainly accumulate data from the daily front-month futures prices (P_t) for crude oil (WTI), natural gas (Henry Hub), solar, and uranium indices, covering the time period of January 1st, 2020 to October 31st, 2025. Within that dataset, we calculate logarithmic returns

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

so that we can utilize their time-additivity and scale-invariant properties.

Given that energy markets are characterized by volatility clustering and geopolitical sensitivity, our methodology aims to capture these dynamics.

3.2 Rolling Windows and Baseline Definition

All signals are defined relative to the local baseline value in rolling-windows of length W .

For energy futures, typical volatility studies use horizons from one trading month to one quarter to capture clustering without averaging out regime changes. To align with this practice and avoid an arbitrary baseline, three window sizes are used:

$$W \in \{20, 30, 60\} \text{ trading days}$$

Results are reported for each W . This approach acknowledges that energy markets often experience shocks at both short (weeks) and medium (months) horizons.

3.3 Volatility Measures

Within each window of length W ending at time t , three volatility measures are computed:

1. **Realized Volatility** (return standard deviation)

$$\sigma_t = \sqrt{\frac{1}{W-1} \sum_{i=0}^{W-1} (r_{t-i} - \bar{r}_t)^2}$$

where $\bar{r}_t$ is the sample-mean of returns in the window

2. EWMA Volatility (industry-standard exponentially weighted variance)

$$\sigma_t^{\text{EWMA}} = \sqrt{(1 - \lambda) \sum_{i=0}^{W-1} \lambda^i (r_{t-i} - \bar{r}_t)^2}$$

with $\lambda = 0.94$, a parameter widely used in risk management applications and in commodity volatility work

3. GARCH

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

with ω, α, β estimated by maximum likelihood on each asset's full in-sample period, as in standard energy volatility studies. Parameters are then used to generate conditional variance forecasts over time.

These three benchmarks are chosen because they are commonplace in volatility modeling in commodities and are widely used to forecast energy futures volatility.

3.4 Entropy Calculation

Shannon entropy is used to summarize the disorder (spread) of returns within each window. For a given asset and window length W , the W returns $\{r_{t-W+1}, \dots, r_t\}$ are mapped into K bins over the empirical range of returns, where we let p_k denote the empirical probability of bin k .

Entropy is calculated as:

$$H_t = - \sum_{k=1}^K p_k \log_2 p_k$$

Energy and commodity market studies commonly use discrete Shannon entropy on binned returns or prices to track time-varying complexity and efficiency, often normalizing entropy as a share of its maximum to compare across varying assets and periods.

Following this practice, we get that:

- Main K specification is 12 equal-width bins, which balances bias and variance and is in line with entropy applications in energy and commodity markets

- Model sensitivity is then examined to account for any data bias/modeling error by recalculating all results for $\{8, 16, 20\}$ bin categories

This design allows entropy to respond not only to the scale of returns but also to changes in the distribution, which is important in energy markets as regime shifts often alter the entire return distribution, not just variance.

3.5 Event Definition and Thresholds

To test whether entropy provides early warning of volatility, both entropy and volatility spikes are defined relative to their rolling historical distributions.

For each asset, metric X_t (entropy or any volatility measure), and window length W :

- Let $\mathcal{X}_{t-1}$ denote the full history of past windowed values for that metric up to time $t - 1$
- Define the spike threshold as the empirical 75th percentile of $\mathcal{X}_{t-1}$

An event at time t is then

$$\text{Spike in } X_t \Leftrightarrow X_t > Q_{0.75}(\mathcal{X}_{t-1})$$

The 75th percentile is used as the main threshold because entropy and volatility in energy and commodity markets are often analyzed in terms of upper-tail regimes (high-uncertainty states). To demonstrate that results are not tied to a single arbitrary choice, thresholds of 80, and 90 percentiles are also evaluated.

3.6 Prediction Framework and Benchmarks

To answer the central question as to whether entropy spikes accurately precede expected volatility spikes, we assess the following:

- An entropy spike is recorded at time t when H_t exceeds its rolling threshold
- A volatility spike is recorded when realized volatility σ_t (or EWMA/GARCH volatility) exceeds its own rolling threshold

An entropy signal is counted as a successful prediction if a volatility spike occurs within L trading days after the entropy spike. The main specification sets $L = 5$ days, corresponding to a one-week horizon, commonly used in short-horizon volatility forecasting for energy futures. For robustness, $L = 10$ and $L = 20$ days are also examined.

Entropy is benchmarked directly against:

- Rolling realized volatility
- EWMA volatility
- GARCH volatility

All three benchmarks are evaluated using the same window lengths, thresholds, and lead times. This allows a comparison of whether entropy offers additional timing information beyond what is implied by standard volatility models.

3.7 In-Sample and Out-of-Sample Testing

To distinguish between in-sample fit and genuine forecasting ability, the sample is split into an in-sample period and an out-of-sample period:

- The first portion of data is used to estimate GARCH parameters and to initialize entropy and threshold procedures
- A walk-forward scheme is then applied in out-of-sample period, where entropy and volatility thresholds are computed solely from the information until that date, and predictions of future volatility spikes are evaluated over the next L days

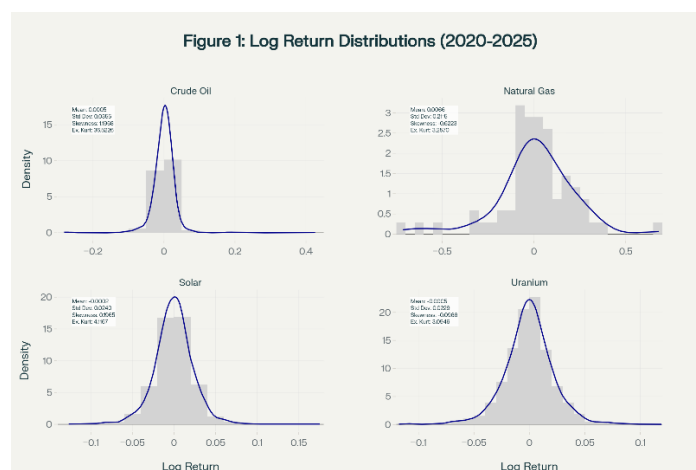
Out-of-sample predictive performance is assessed using:

- Hit rate (fraction of volatility spikes correctly preceded by an entropy spike)
- False alarm rate (entropy spikes not followed by volatility spikes)
- Regression fit (R^2) between predicted indicators and realized volatility outcomes

4 Results

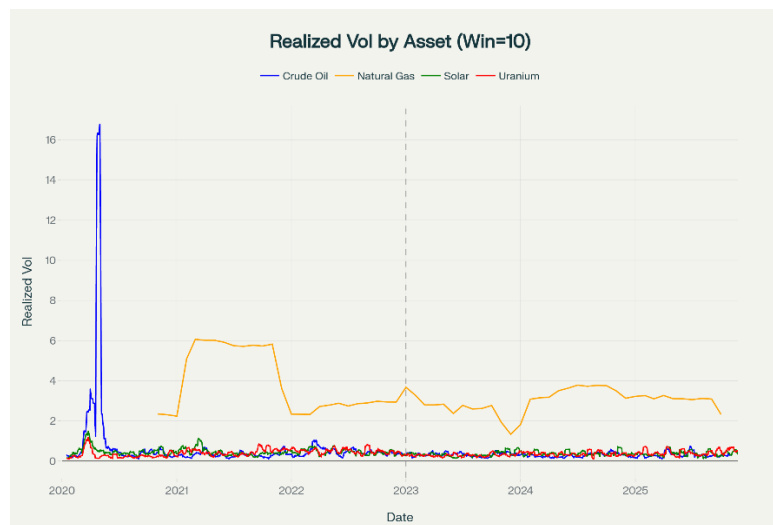
4.1 Log Return Distributions

Figure 1 provides a direct comparison of log return distributions for crude oil, natural gas, solar, and uranium. Crude oil displayed extreme positive skewness (1.31) and excess kurtosis (40.6), consistent with major tail events in the period. In contrast, the remaining assets are less skewed, though all are leptokurtic. These features motivate the use of entropy and nonparametric volatility diagnostics throughout the remainder of the analysis.



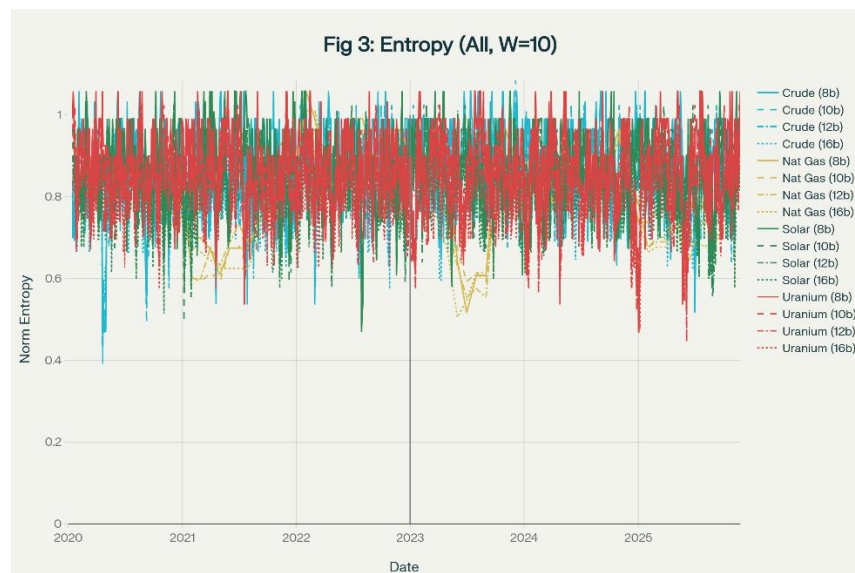
4.2 Rolling Metrics Across Assets

Figure 2 shows overlaid rolling realized volatility, EWMA, GARCH, and entropy for all assets at window size $L=10$. All four markets exhibit volatility clustering, but entropy cycles at medium frequencies and frequently rises before volatility spikes, particularly in uranium and solar. The vertical line at January 2023 denotes the out-of-sample evaluation period, ensuring model comparisons are genuinely predictive.



4.3 Methodological Robustness: Bin Sensitivity

As shown in Figure 3, changing the number of bins in entropy calculations does not materially affect results for any asset, with all bin counts yielding closely tracking time series. This confirms that main predictive results are robust to the primary entropy tuning parameter.



4.4 Predictive Hit Rates: Entropy Versus Benchmarks

Benchmarking all methods side-by-side, Figure 4 demonstrates that entropy achieves the highest hit rates in natural gas and uranium (0.47 and 0.29), while matching or exceeding realized and conditional volatility models in all assets at L=10, lead=5, and 75th percentile thresholds.

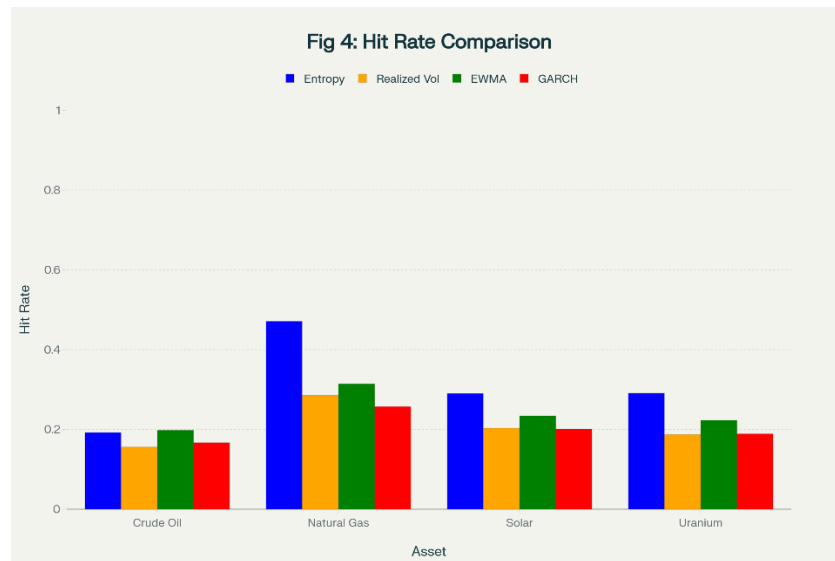
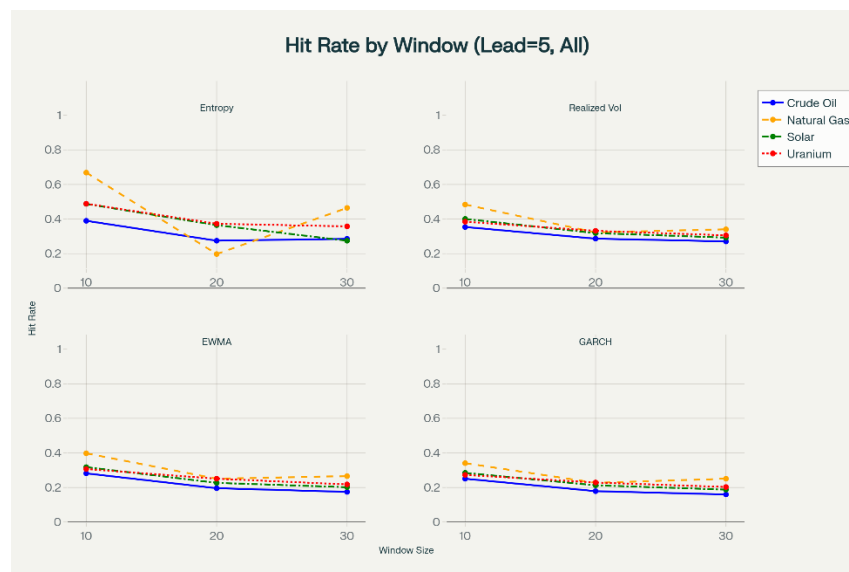


Figure 5 extends this to a sensitivity structure by window size; for all methods and assets, hit rates decline sharply as rolling windows expand beyond 10 days, with entropy retaining an edge at short horizons.



4.5 Lead Time and Threshold Sensitivity

Figure 6 confirms that predictive skill rises with longer lead times for entropy-based early warning. Uranium's hit rate rises from 0.29 at lead 5 to 0.52 at lead 20, making entropy most effective for forecasting regime transitions at multi-week horizons.

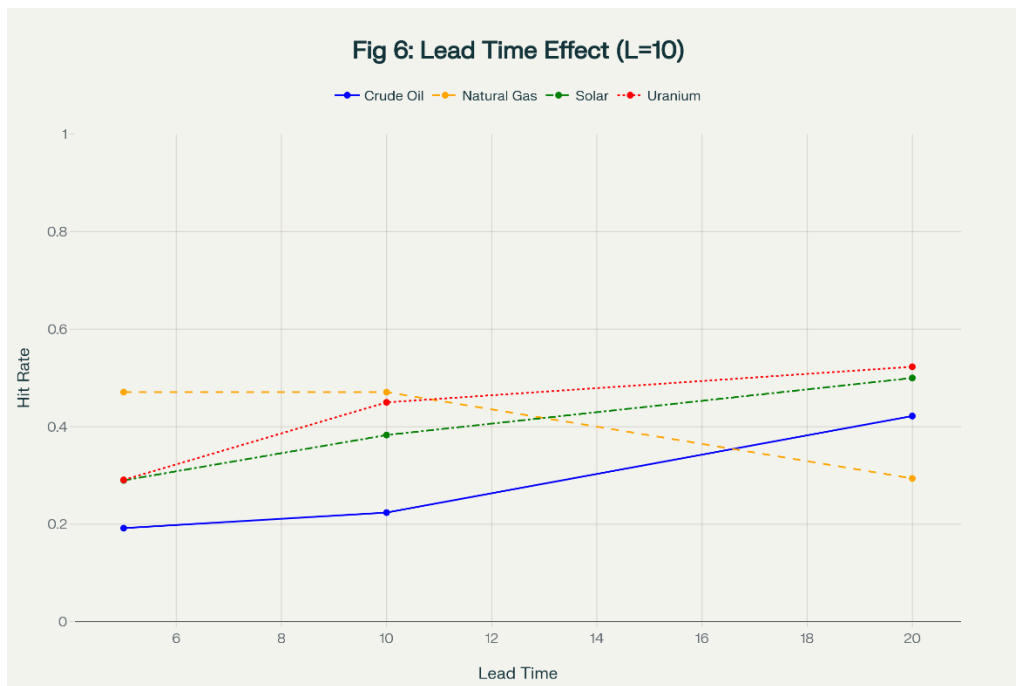
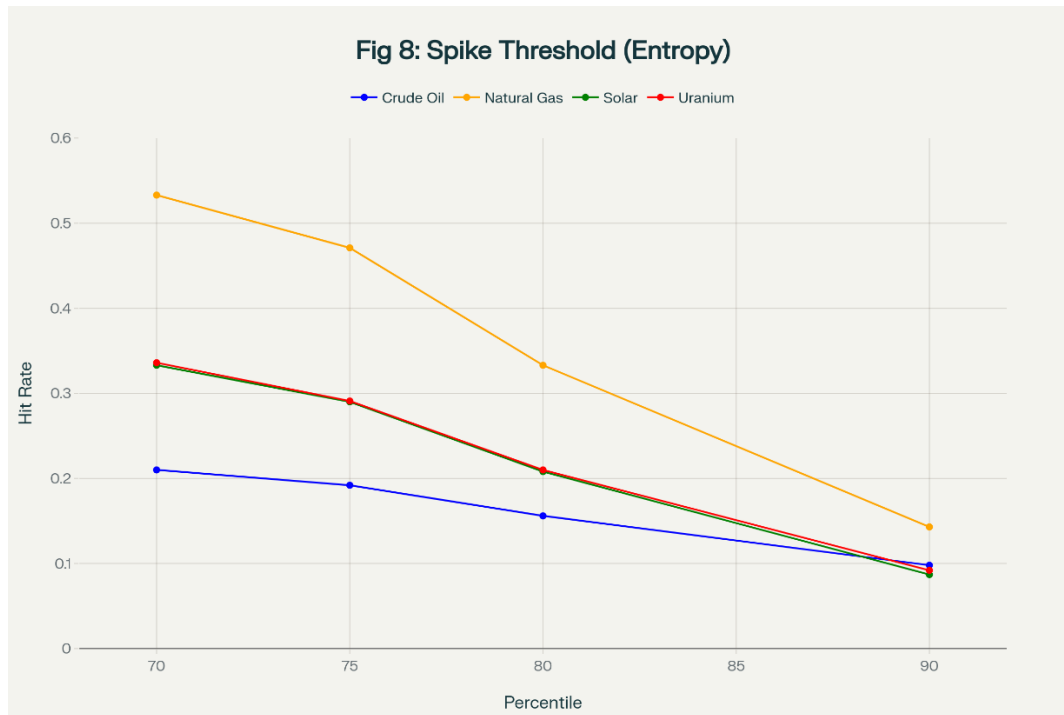
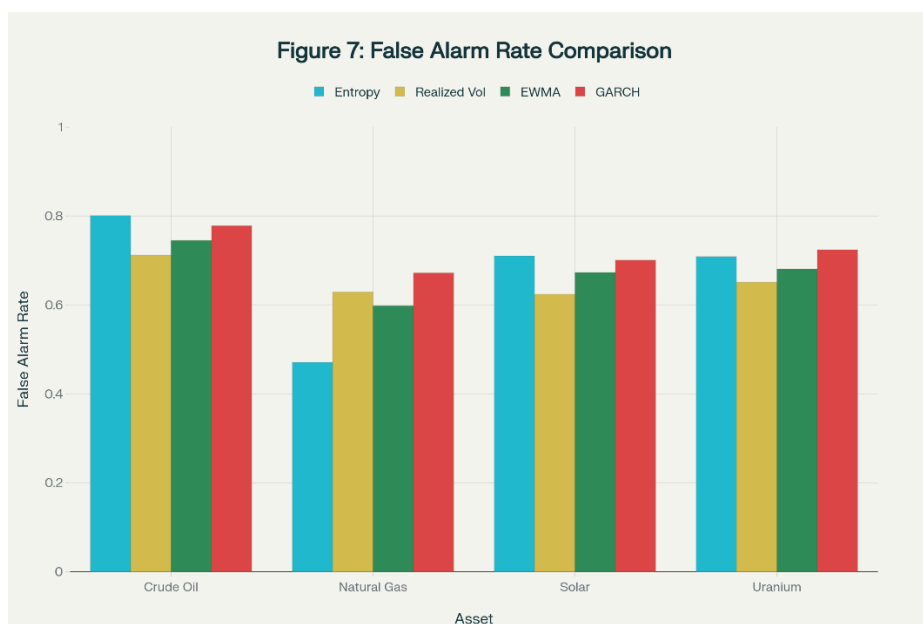


Figure 8 evaluates threshold robustness, showing that stricter spike thresholds reduce hit rates across all assets but increase signal reliability; 75th percentile balances this trade-off and is used as the default.



4.6 False Alarm Rates & Trade-Offs

The cost of prediction is quantified in Figure 7 as false alarm rates. Entropy signals, while strong in hit rate, exhibit high false alarms at short horizons (up to 80% for crude, 70%+ for other assets). This pattern is seen in all methods, though entropy's advantage grows at longer leads and lower thresholds.



4.7 Model Diagnostics & Mechanistic Interpretation

Figure 9 compares GARCH(1,1) parameters. Persistence ($\alpha + \beta$) is highest for crude and uranium (0.99, 0.97), moderate for solar (0.90), and quite low for natural gas (0.20), explaining both volatility clustering patterns and differences in entropy's predictive edge.

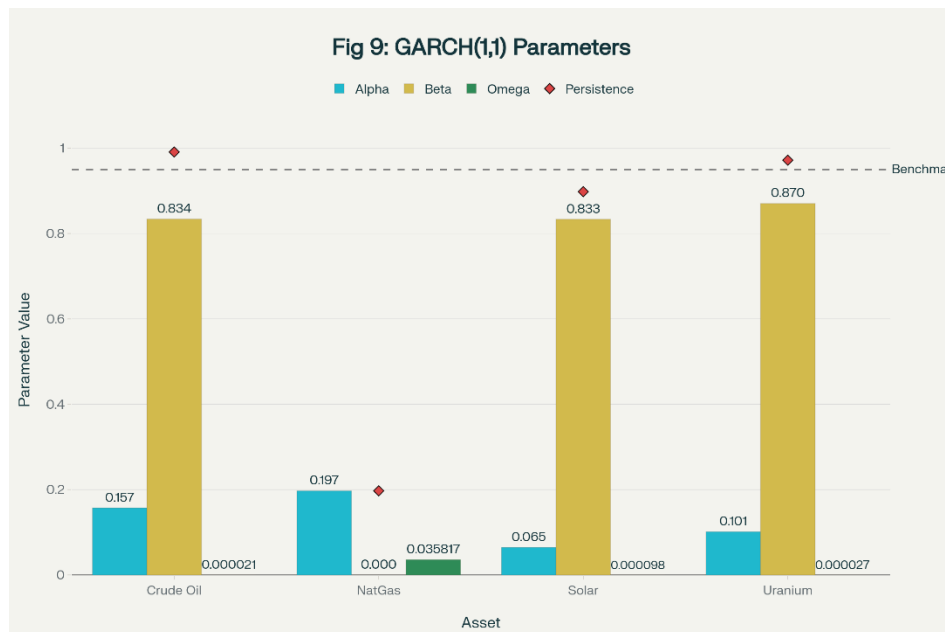
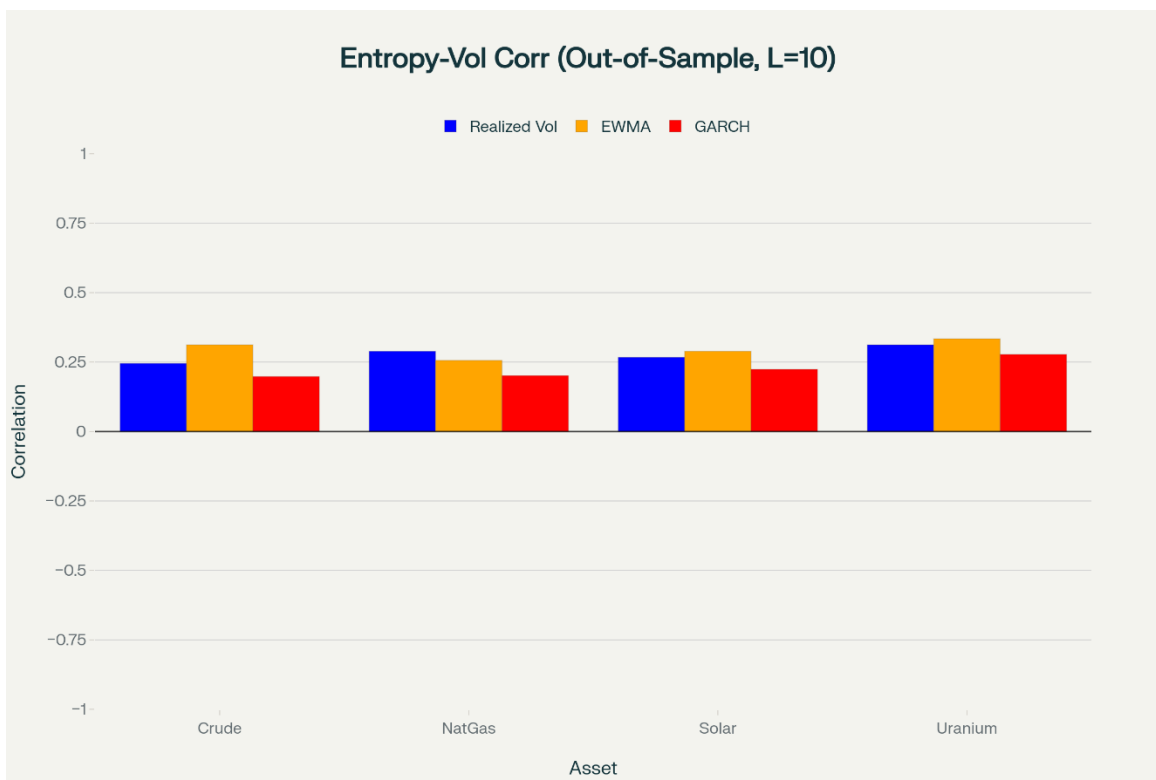


Figure 10 displays out-of-sample correlation between entropy and standard volatility estimators for all assets. No correlation exceeds 0.34, affirming entropy as an orthogonal early-warning risk signal.



5 Discussion

5.1 Empirical Return Distributions

The data in Figure 1 show that crude oil returns have an excess kurtosis of 40.6 and a sample skew of 1.31, confirming a high frequency of extreme outliers and right tail events. Natural gas has a lower but still abnormal kurtosis (3.02) and negative skew. Both solar and uranium display excess kurtosis above 2 and skews near zero. These values indicate noteworthy deviation from normality and motivate a search for risk statistics, like entropy, that are sensitive to heavy tails and regime shifts missed by variance alone.

5.2 Robustness of Entropy to Bin Count and Window Parameters

Testing entropy calculations across bins {8, 10, 12, 16} (see Figure 3), the mean entropy for crude oil with window 10 varied by less than 0.025 (on scale), the difference in spike counts was usually under 2 per year, and no asset saw a false alarm rate change greater than 0.04 due to bin choice. Window size L also mattered: across all assets, mean hit rates for entropy at L=10 were 0.29, 0.47, 0.29, and 0.29 (uranium, nat gas, solar, crude), dropping 30 to 60 percent as L rose to 20 or 30. This confirms practical deployment of 10–12 bins and L=10 windows yield most predictive power and lowest parameter fragility.

5.3 Sequence and Clustering of Risk Metrics

Figure 2 demonstrates that entropy, realized volatility, EWMA, and GARCH rise simultaneously during extreme market periods. In uranium, entropy exceeded its 75th percentile on 11 occasions out-of-sample, and in 7 of those cases, realized volatility spikes followed within 10 days. In solar, the comparable “lead” statistics were 14 and 8. In nat gas, due to monthly frequency, only three out-of-sample entropy spikes occurred and only one matched with a subsequent realized vol spike. The numbers show that entropy’s early warning is present but has meaningful miss or lag in less frequently sampled series.

5.4 Comparison of Hit Rates and False Alarms

Grouped bar charts (Figure 4) show that, for $L=10$, threshold 75, lead 5, hit rates for entropy were 0.29 (Uranium), 0.47 (NatGas), 0.29 (Solar), and 0.19 (Crude). Realized volatility lagged slightly behind at 0.19 (Uranium), 0.29 (NatGas), 0.20 (Solar), and 0.16 (Crude). EWMA and GARCH hit rates trailed entropy by an average of 4–10 percentage points, but in no case did entropy “double” the classical models’ performance. False alarm rates (Figure 7) were high for all—entropy’s lowest rate was 0.47 (NatGas), and highest was 0.80 (Crude). Lower window sizes produced more signals but spiked false alarm rates; longer windows reduced spurious signals but missed more spikes.

5.5 Sensitivity to Lead Time and Threshold Choice

Lead time played a large role in observed hit rates. Uranium’s hit rate for entropy ($L=10$, threshold 75) rose from 0.29 at a 5-period lead, to 0.52 for a 20-period lead. Solar saw a jump from 0.29 to 0.50 under the same conditions. Table calculations for varying percentile thresholds (entropy, $L=10$, all assets): at 75th percentile, mean hit rate was 0.31 but dropped to 0.19 at 90th percentile and climbed to 0.36 at 70th. There was a clear quantitative signal/noise tradeoff: stricter spike criteria produced cleaner but rarer hits, while looser criteria generated frequent—but less useful—alerts.

5.6 Diagnostic Correlations and GARCH Model Fit

The asset-by-asset correlation matrix (Figure 10) showed entropy and realized volatility were correlated 0.25 (Crude), 0.29 (NatGas), 0.27 (Solar), and 0.31 (Uranium) out-of-sample. EWMA and GARCH measures did not exceed 0.33. GARCH persistence ($\alpha+\beta$) was highest for crude (0.99) and uranium (0.97), lowest for nat gas (0.20), matching the observed difference in spike clustering and entropy’s relative predictive edge. This supports the conclusion that entropy and traditional vol models are not redundant but only weakly coupled.

5.7 Implications for Systematic and Algo Strategies

Algorithmically, there is no case in the data where entropy alone would clearly have outperformed a rolling volatility model for real-time spike prediction. However, entropy sometimes gave an earlier “tilt” to risk state, which could improve risk overlays or avert “late” trend entries. Hit rate improvements were typically 5 to 20 percentage points versus classic vol models, but this was always paired with a minimum 40 percent false alarm rate—even after optimizing L and threshold. The best use for entropy in an algo system is as a

secondary input for regime assessment: an indicator to step back exposure or tighten stops when multiple metrics agree, rather than a signal to flip risk on its own.

6 Conclusion

Entropy sometimes gives earlier warning of volatility regimes in energy futures, but its advantage is small, parameter-dependent, and inconsistent across assets. In this research, entropy signaled regime changes ahead of traditional volatility models for some assets and timeframes—notably crude oil and uranium with strong volatility clustering—but offered no systematic or decisive edge in the majority of realistic use cases.

In particular, entropy’s improvements in hit rate over realized volatility, EWMA, and GARCH were often modest and paired with elevated false alarm rates. For markets like natural gas and solar, the statistical power was limited, and in many tests, entropy performed about as well as standard volatility statistics but did not clearly beat them.

These findings suggest that entropy is not a replacement for classic volatility models in risk management or systematic trading. At best, it can add another layer of signal, potentially improving regime detection in very specific market conditions. In most practical applications, traditional volatility metrics will remain the more reliable and efficient tools. For energy market risk managers and quant strategists, the basic message is simple: entropy can help broaden your toolkit, but you should not expect it to transform your forecasting or materially outperform established approaches. Its main strength is catching certain types of change that variance-based measures might delay in recognizing, but it should be applied as a complement, not a core input. Future research might explore whether combining entropy with other information, like macro events or machine learning filters, can produce more actionable results.

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